

7.7 Solution: (a) From

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t},$$

when there is no external current, we have

$$\nabla \times (\nabla \times \mathbf{E}) = -\nabla \times \frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{H}) = -\mu_0 \frac{\partial^2 \mathbf{D}}{\partial t^2}.$$

Assuming $\mathbf{E} \propto e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t}$ and also $\mathbf{D} \propto e^{i\mathbf{k} \cdot \mathbf{x} - i\omega t}$, the above equation becomes

$$i\mathbf{k} \times (i\mathbf{k} \times \mathbf{E}) = \mu_0 \omega^2 \mathbf{D},$$

or equivalently,

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{E}) + \mu_0 \omega^2 \mathbf{D} = 0.$$

(b) Using the identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$, we can express the wave equation as

$$\mathbf{k}(\mathbf{k} \cdot \mathbf{E}) - k^2 \mathbf{E} + \mu_0 \omega^2 \mathbf{D} = 0.$$

Let $\mathbf{k} = k\mathbf{n}$, then we will have

$$\mathbf{n}(\mathbf{n} \cdot \mathbf{E}) - \mathbf{E} + \mu_0 v^2 \mathbf{D} = 0,$$

with $v = \omega/k$. Since we have chosen the coordinate axes to coincide with the principal axes, $D_i = \epsilon_i E_i$, $i = 1, 2, 3$, the component-wise form of the above equation is

$$n_i(\mathbf{n} \cdot \mathbf{E}) - E_i + \frac{v^2}{v_i^2} E_i = 0, \quad (1)$$

where $v_i^2 = 1/\mu_0 \epsilon_i$, or

$$E_i = \frac{n_i(\mathbf{n} \cdot \mathbf{E})}{1 - v^2/v_i^2}.$$

Multiply both sides by n_i and sum over the axes, we will obtain

$$(\mathbf{n} \cdot \mathbf{E}) = \sum_{i=1}^3 \frac{n_i^2}{1 - v^2/v_i^2} (\mathbf{n} \cdot \mathbf{E}),$$

or

$$\sum_{i=1}^3 \frac{n_i^2}{1 - v^2/v_i^2} = 1 = \sum_{i=1}^3 n_i^2.$$

Therefore,

$$\sum_{i=1}^3 n_i^2 \left(\frac{1}{1 - v^2/v_i^2} - 1 \right) = -v^2 \sum_{i=1}^3 \frac{n_i^2}{v^2 - v_i^2} = 0,$$

which finally leads to, for $v \neq 0$, the Fresnel equation,

$$\sum_{i=1}^3 \frac{n_i^2}{v^2 - v_i^2} = 0. \quad (2)$$

Of course, $v = 0$ is a valid solution to the original equation. However, this is not propagating and thus can be dropped. The propagation modes can be determined by Eq. (2), which is a quadratic equation in v^2 and has two different nonzero solutions.

(c) The component-wise equation (1) can also be written as

$$n_i \sum_{j=1}^3 n_j \frac{D_j}{\epsilon_j} - \frac{D_i}{\epsilon_i} + \mu_0 v^2 D_i = 0,$$

or

$$n_i \sum_{j=1}^3 n_j v_j^2 D_j - v_i^2 D_i + v^2 D_i = 0. \quad (3)$$

Multiply both sides by n_i and sum over the principal axes, noticing that $\sum_{i=1}^3 n_i^2 = 1$, we have

$$\sum_{i=1}^3 n_i^2 \cdot \sum_{j=1}^3 n_j v_j^2 D_j - \sum_{i=1}^3 n_i v_i^2 D_i + v^2 \sum_{i=1}^3 n_i D_i = v^2 \sum_{i=1}^3 n_i D_i = 0.$$

For propagation modes, $v \neq 0$. Therefore,

$$\sum_{i=1}^3 n_i D_i = \mathbf{n} \cdot \mathbf{D} = 0,$$

for propagation modes, *i.e.*, $\mathbf{D}$ is transverse. Denote the two propagation modes as $\mathbf{D}_a$ and $\mathbf{D}_b$, Eq. (3) can be written as

$$n_i \sum_{j=1}^3 n_j v_j^2 D_{a,j} - v_i^2 D_{a,i} + v_a^2 D_{a,i} = 0, \quad n_i \sum_{j=1}^3 n_j v_j^2 D_{b,j} - v_i^2 D_{b,i} + v_b^2 D_{b,i} = 0,$$

where v_a and v_b are the associated propagation velocity, distinct solutions to the Fresnel equation. Multiply the first equation by $D_{b,i}$ and sum over the principal axes, and second equation by $D_{a,i}$ and take the sum, we will have

$$(\mathbf{n} \cdot \mathbf{D}_b) \cdot \sum_{j=1}^3 n_j v_j^2 D_{a,j} - \sum_{i=1}^3 v_i^2 D_{a,i} D_{b,i} + v_a^2 \mathbf{D}_a \cdot \mathbf{D}_b = 0,$$

and

$$(\mathbf{n} \cdot \mathbf{D}_a) \cdot \sum_{j=1}^3 n_j v_j^2 D_{a,j} - \sum_{i=1}^3 v_i^2 D_{a,i} D_{b,i} + v_b^2 \mathbf{D}_a \cdot \mathbf{D}_b = 0,$$

Take the difference between the two equations and notice the transversality of $\mathbf{D}$, we will arrive at

$$(v_a^2 - v_b^2) \mathbf{D}_a \cdot \mathbf{D}_b = 0.$$

Since the two distinct propagation modes have different velocities, we must have

$$\mathbf{D}_a \cdot \mathbf{D}_b = 0.$$